A Search for CPT and Lorentz Invariance Violations Using Neutrinos at the NOνA Near Detector

Progress Update

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NOvA Collaboration Meeting
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Sidereal Time

- A **solar day** is the time it takes for the Sun to return to the same position in the sky
- A **sidereal day** is the length of one rotation about the Earth's axis
- The sidereal day is approximately four minutes shorter than the solar day

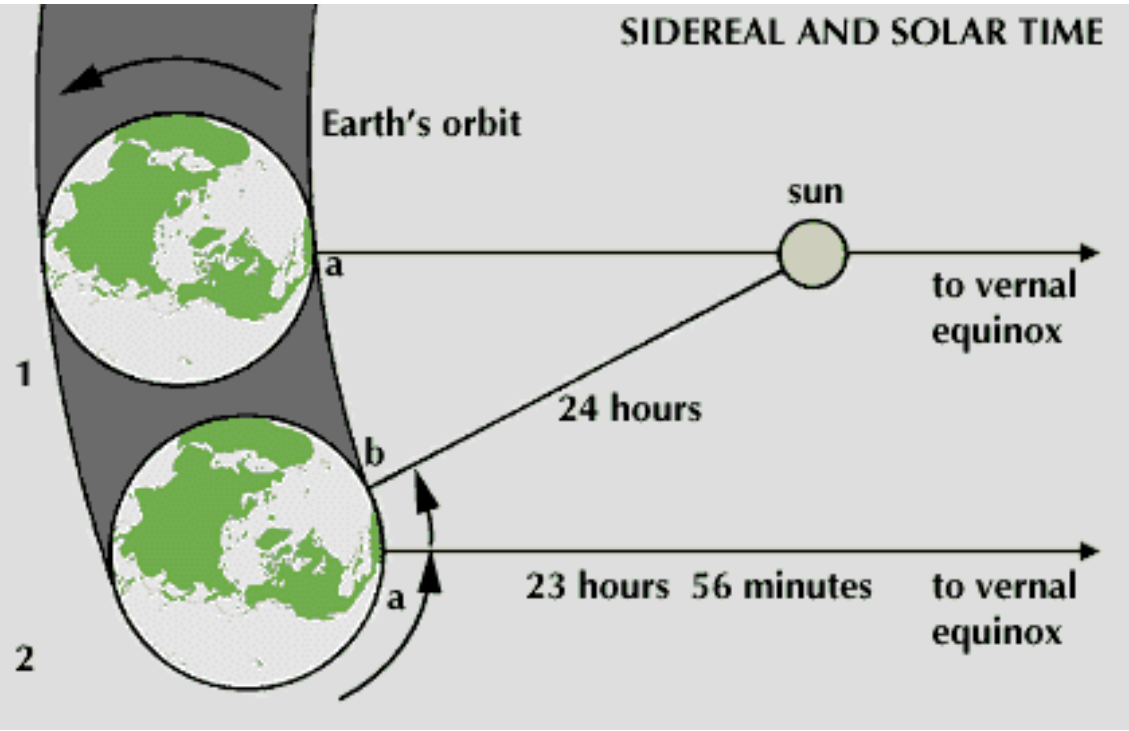


Image adapted from www.schoolphysics.co.uk (Solar and Sidereal Days).

Sidereal Modulation

- If LV exists, we will observe these effects as sidereal modulations on the neutrino event rate
- Rotation of the NuMI beam in the background field allows access to sidereal modulation
- The directional cosines describe the orientation of the NuMI beam in the background:

$$\begin{pmatrix} \hat{N}^X \\ \hat{N}^Y \\ \hat{N}^Z \end{pmatrix} = \begin{pmatrix} \cos\chi \sin\theta \cos\phi + \sin\chi \cos\theta \\ \sin\theta \sin\phi \\ -\sin\chi \sin\theta \cos\phi + \cos\chi \cos\theta \end{pmatrix}$$

Colatitude Azimuth Zenith

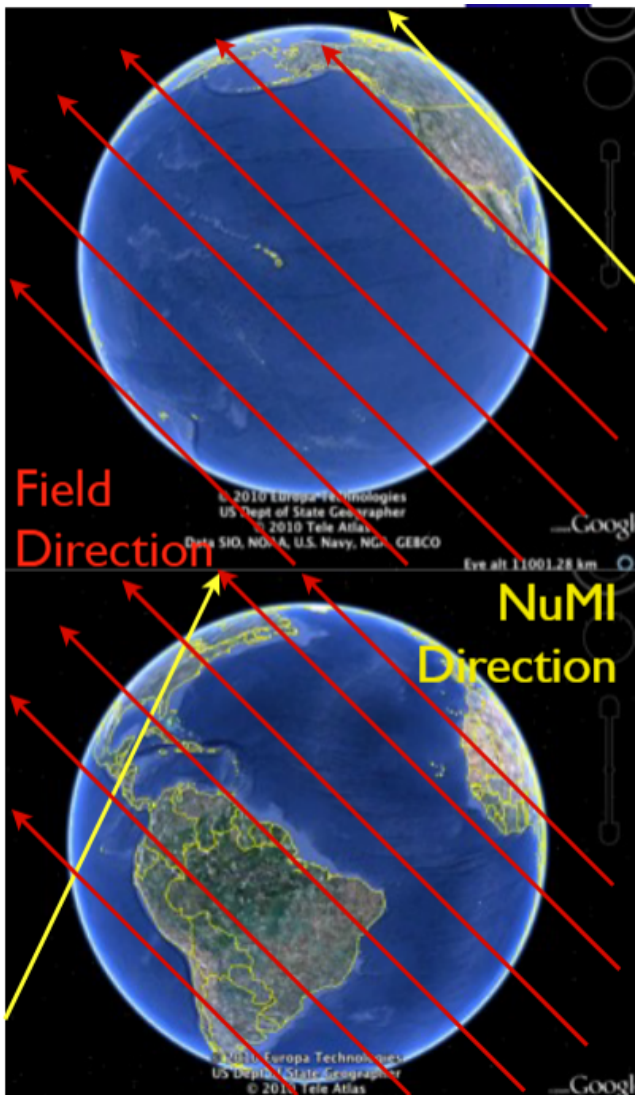


Image courtesy of Prof. Brian Rebel, UW–Madison ν-group, 2011.

Event–Spill Matching Framework

Goal:

- Create a data structure that links each detector event to its corresponding beam spill using shared **(run, subrun, event)**.

Input Files:

- Detector events (**run**, **subrun**, **event**, **selection**, **unixTime**, cycle, slice, cvnScore)
- Beam spills (**run**, **subrun**, **event**, **spillPOT**, **spillTimeSec**, cycle, spillTimeNSec, isMC)

Purpose:

- Validate detector–beam consistency for Lorentz-violation analysis

Matching Process (data + MC):

- Convert unixTime to local sidereal phase (LSP)
- Build unique IDs for every event and spill $\rightarrow$ run:subrun:event
- Match events and spills by ID
- Record events with no matching spill ("unmatched events") and spills with no matching events ("unmatched spills") in text files
- Save merged matching results into a single ROOT file

Spill-Matching Diagnostics: Data vs Monte Carlo

Data Sample:

- Total events: 6.25 M
- Total spills: 5.91 M
- Unmatched events: 342 k (5.5 %)
- Unmatched Spills: 0

Monte Carlo Sample:

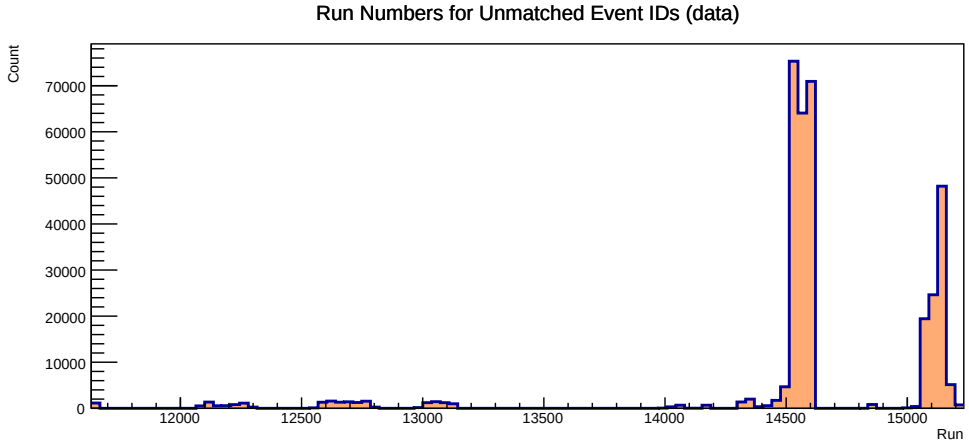
- Total events: 3.37 M
- All events matched perfectly $\rightarrow$ 0 unmatched events or spills

Interpretation:

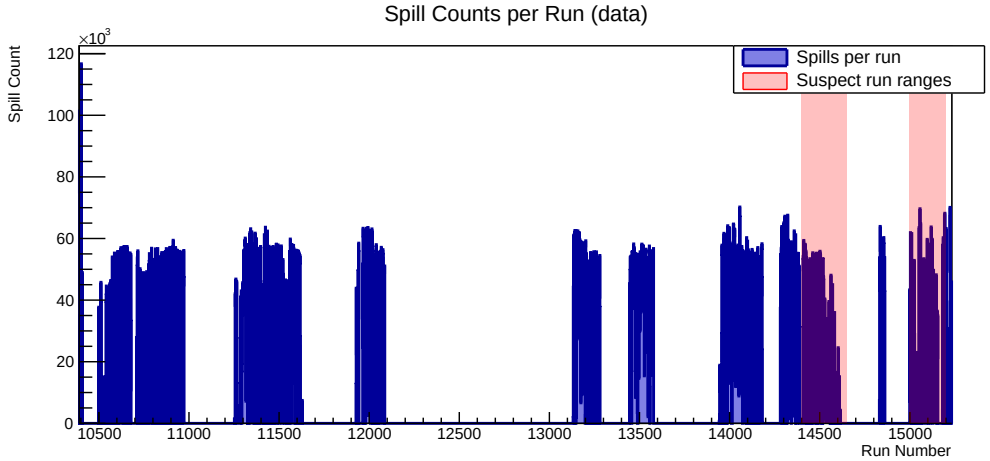
- MC result confirms that the matching algorithm works correctly
- Data sample contains a significant fraction of unmatched events (5.5 %) — this suggests a possible issue with the data

Unmatched Data Event Distribution and Spill Diagnostics

Unmatched Events vs Run Number



Spill Counts vs Run Number (highlighted suspect ranges)



Suspect regions (large amount of unmatched events):

- Runs 14400–14650 → Jan–Jul 2022
- Runs 15000–15200 → Feb–Jun 2023

Notes:

- Spill distribution appears consistent in suspect regions

Unmatched Data Events Investigation

Comparing Suspect vs Normal Runs:

Metric	Normal Runs	Suspect Runs	Ratio
Mean unmatched events/run	41	1,319	3217.1%
Mean spills/run	27,000	20,000	72.4%

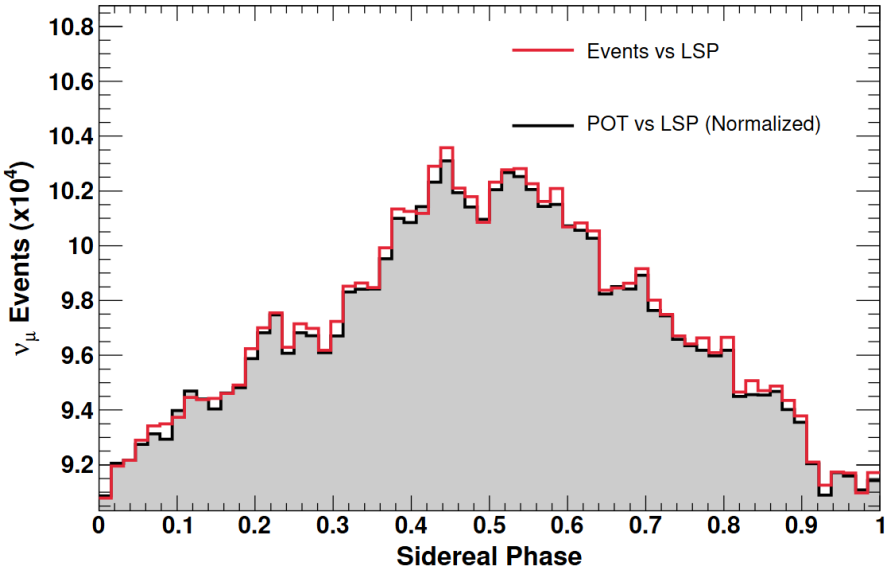
- 3217.1% ratio for unmatched events suggests possible mismatched spill IDs
- Spill density is only ∼28% lower — likely no spill data lost

Unmatched Event Proximity to Nearby Spills:

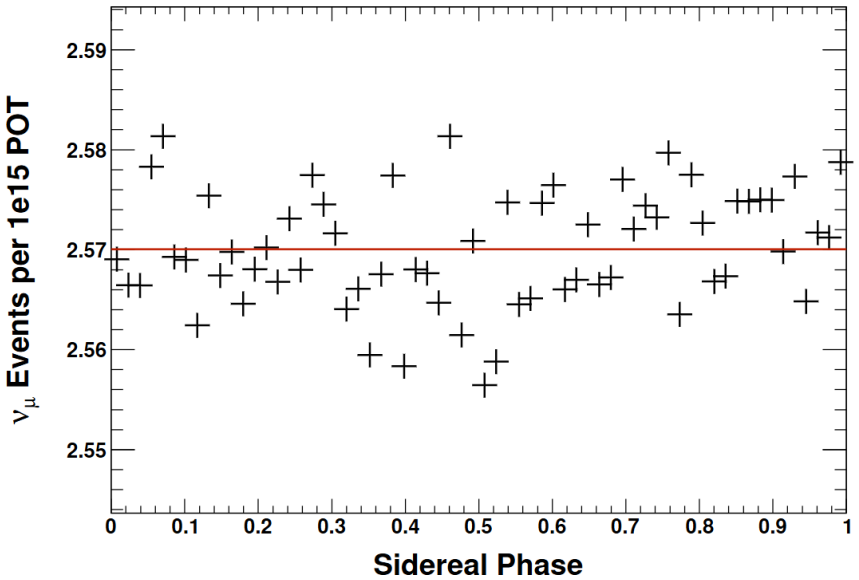
- ∼315k unmatched data events found in suspect runs
- ∼29% within 1 s of a recorded spill, ∼36% within 2 s, ∼48% within 5 s, ∼58% within 10 s
- Median $\Delta t \approx 1$ s
- 5–95% span: -169 s to $+2608$ s $\rightarrow$ most unmatched events occurred within ∼3 min before to ∼43 min after the nearest spill
- Positive offset may suggest timing or indexing drift

Simulated Experiment

- MC events are assigned the timestamp of their beam spill
- Assign each simulated spill a random timestamp from POT vs LSP

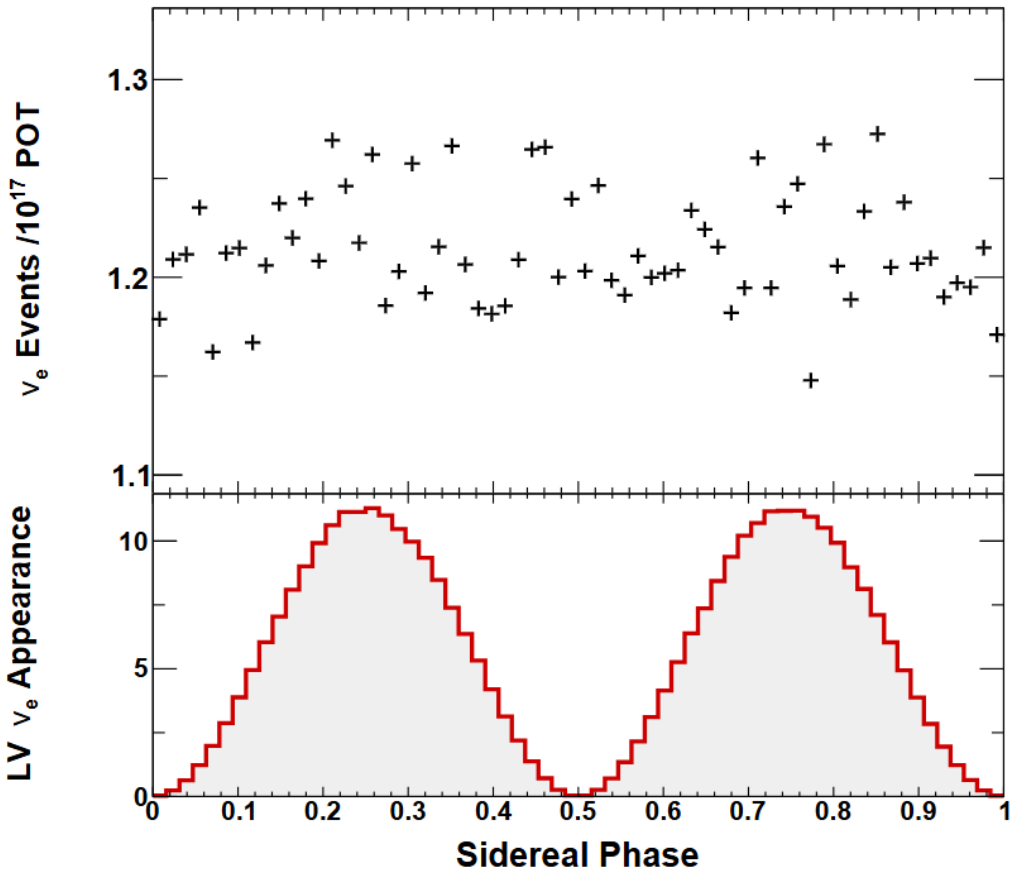


- Divide MC events vs LSP by POT vs LSP
- This simulated experiment has no LV modulation



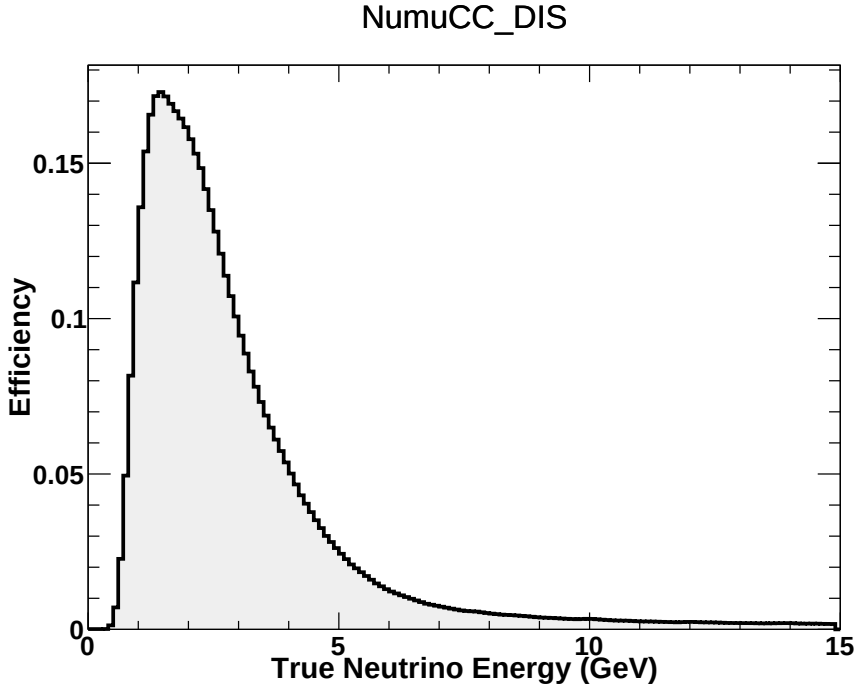
LV Event Rate

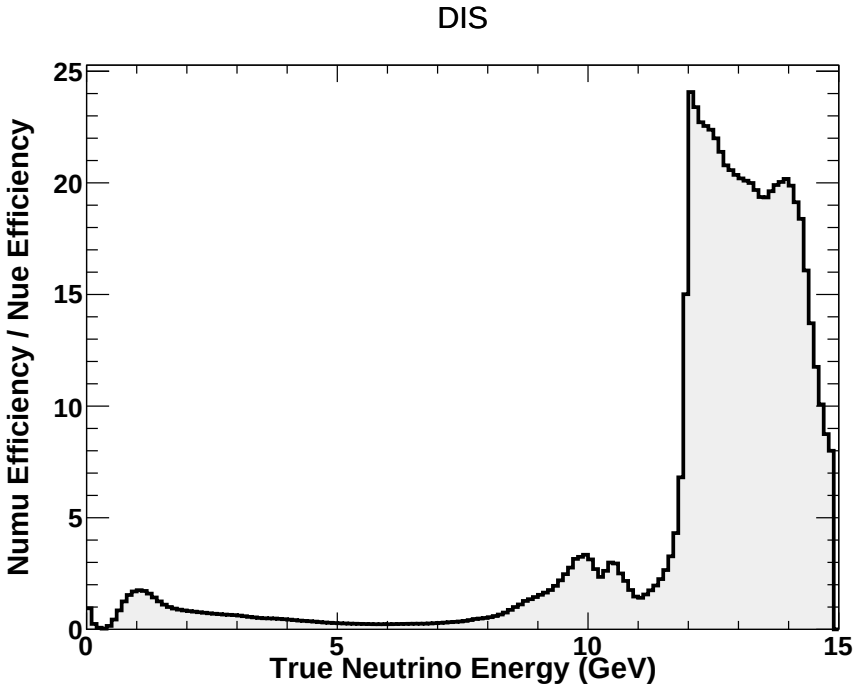
- We may gain greater sensitivity to LV by considering the ν_e channel
- FFT can identify small modulation in the event rate



ND Efficiency

- LV could cause $\nu_\mu \leftrightarrow \nu_e$ transitions
- Account for different efficiency between channels
- Consider fraction of all data events which pass reconstruction and standard selection cuts*





ND Efficiency histograms courtesy of Jozef Trokan-Tenorio of the UW-Madison ν-group.

Next Steps

Current Progress

- LV physics integrated
- Collected MC events
- Using `run/subrun/event` for event lookup

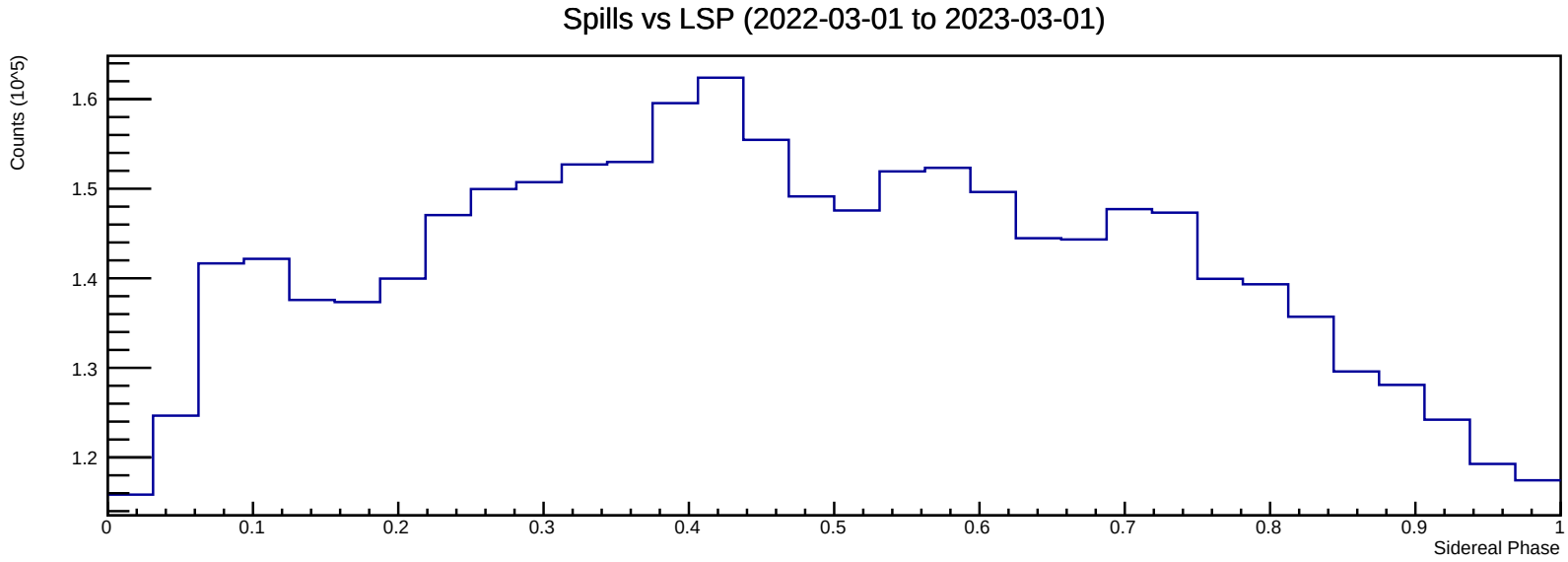
Next Steps

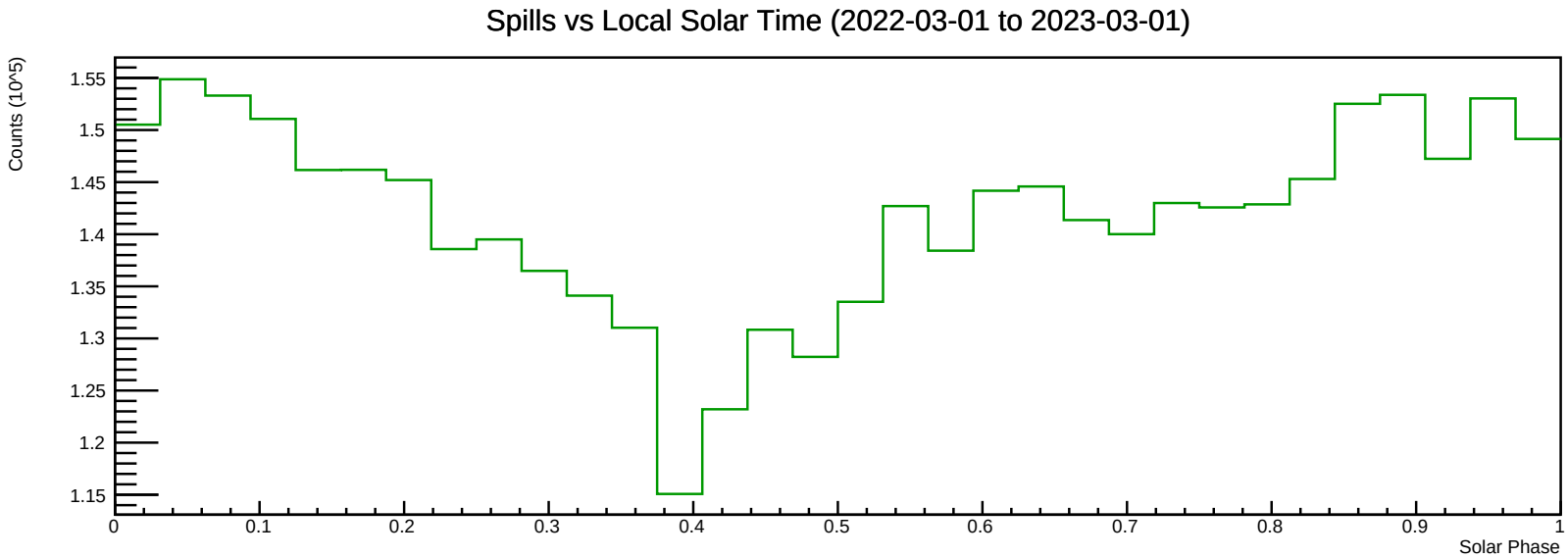
- Investigate spill/event matching
- Integrate event selection efficiency weighting for $\nu_\mu \leftrightarrow \nu_e$ transitions

Thank you for listening!

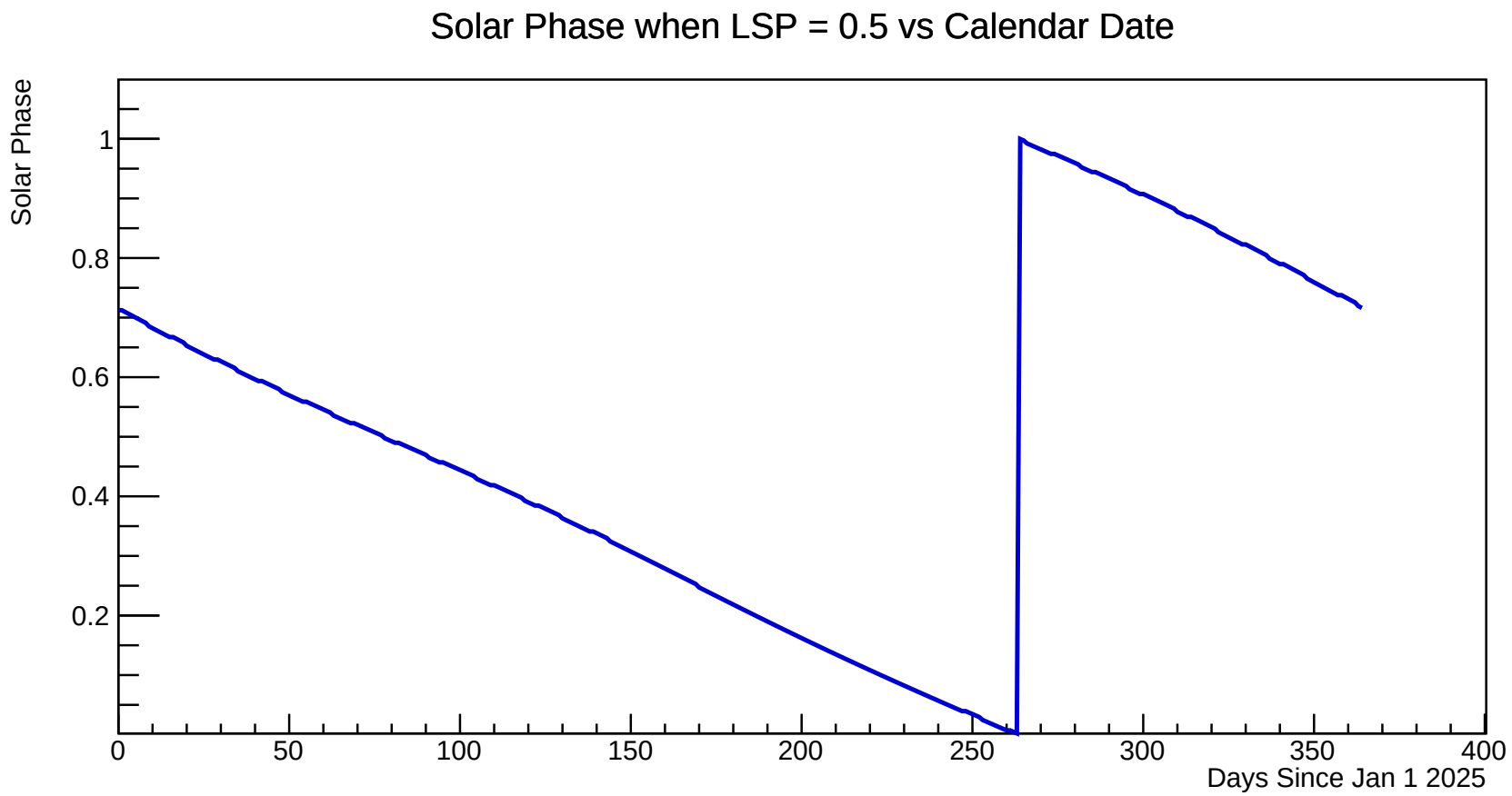
We gratefully acknowledge the many contributions from the members of the UW-Madison ν-group, NOνA Collaboration, and FermiLab staff.

Sidereal vs. Solar Time





Sidereal vs. Solar Time Cont.



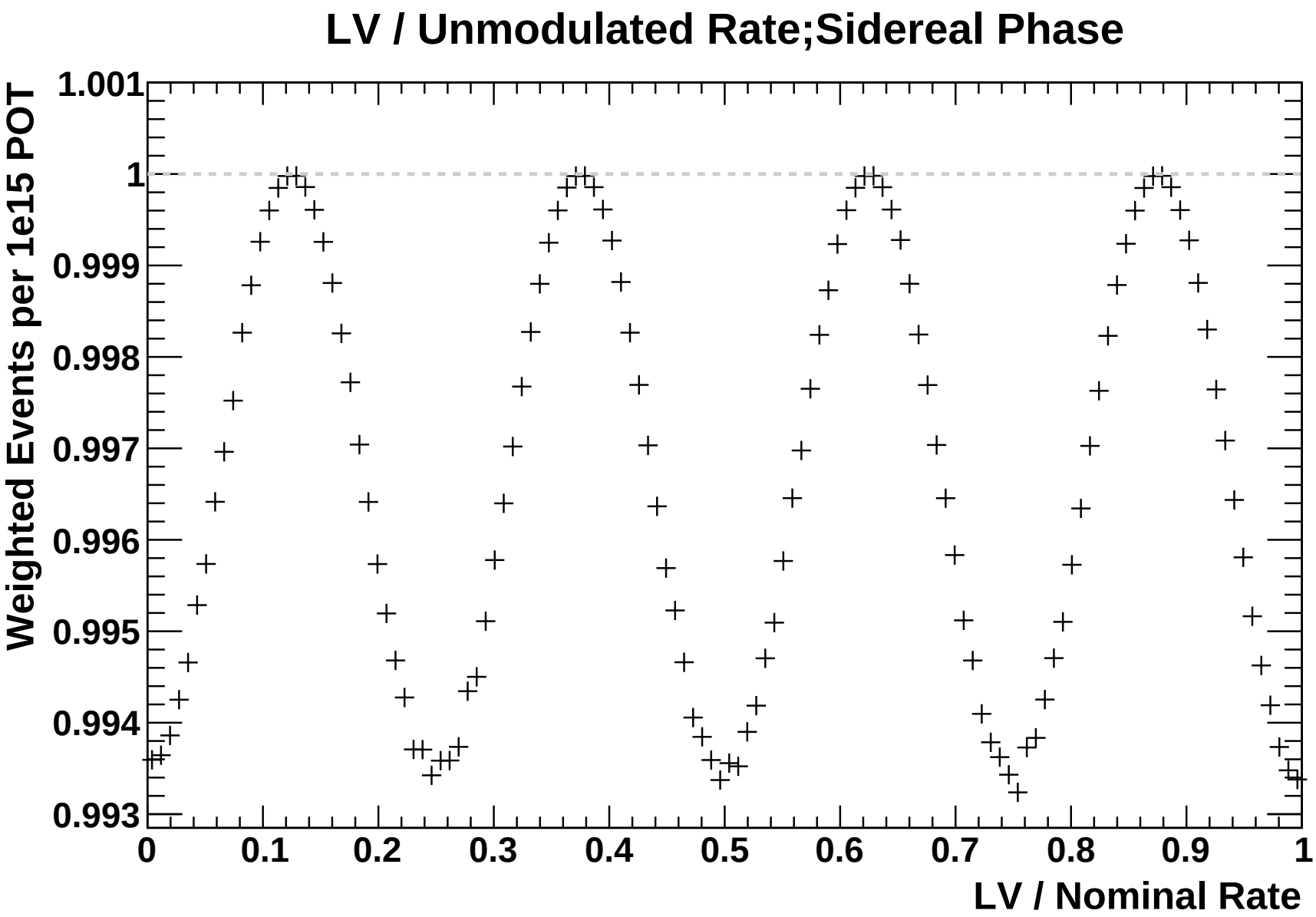
SME Intermediate Calculations

$$
\begin{aligned}
(\mathcal{A}_s^{(0)})_{ab} &= \hat{N}^Y (a_L)^X_{ab} - \hat{N}^X (a_L)^Y_{ab}, \\
(\mathcal{A}_s^{(1)})_{ab} &= -2\hat{N}^Y (c_L)^{TX}_{ab} + 2\hat{N}^X (c_L)^{TY}_{ab} \\
&\quad + 2\hat{N}^Y \hat{N}^Z (c_L)^{XZ}_{ab} - 2\hat{N}^X \hat{N}^Z (c_L)^{YZ}_{ab}, \\
(\mathcal{A}_c^{(0)})_{ab} &= -\hat{N}^X (a_L)^X_{ab} - \hat{N}^Y (a_L)^Y_{ab}, \\
(\mathcal{A}_c^{(1)})_{ab} &= 2\hat{N}^X (c_L)^{TX}_{ab} + 2\hat{N}^Y (c_L)^{TY}_{ab} \\
&\quad - 2\hat{N}^X \hat{N}^Z (c_L)^{XZ}_{ab} - 2\hat{N}^Y \hat{N}^Z (c_L)^{YZ}_{ab},
\end{aligned}
$$

$$
\begin{aligned}
(\mathcal{B}_s^{(1)})_{ab} &= \hat{N}^X \hat{N}^Y \big((c_L)^{XX}_{ab} - (c_L)^{YY}_{ab}\big) \\
&\quad - \big(\hat{N}^X \hat{N}^X - \hat{N}^Y \hat{N}^Y\big)(c_L)^{XY}_{ab}, \\
(\mathcal{B}_c^{(1)})_{ab} &= -\tfrac{1}{2}\big(\hat{N}^X \hat{N}^X - \hat{N}^Y \hat{N}^Y\big)\big((c_L)^{XX}_{ab} - (c_L)^{YY}_{ab}\big) \\
&\quad - 2\hat{N}^X \hat{N}^Y (c_L)^{XY}_{ab}.
\end{aligned}
$$

$$
\begin{aligned}
(\mathcal{C})_{ab} &= (\mathcal{C}^{(0)})_{ab} + E(\mathcal{C}^{(1)})_{ab}, \\
(\mathcal{A}_s)_{ab} &= (\mathcal{A}_s^{(0)})_{ab} + E(\mathcal{A}_s^{(1)})_{ab}, \\
(\mathcal{A}_c)_{ab} &= (\mathcal{A}_c^{(0)})_{ab} + E(\mathcal{A}_c^{(1)})_{ab}, \\
(\mathcal{B}_s)_{ab} &= E(\mathcal{B}_s^{(1)})_{ab}, \quad (\mathcal{B}_c)_{ab} = E(\mathcal{B}_c^{(1)})_{ab}
\end{aligned}
$$

LV / Unmodulated Ratio



SME Hamiltonian Expansion

- $\mathcal{H}_{\text{eff}}$ expansion in terms of sidereal time to first order*

$$\mathcal{H}_{\text{eff}} = A_s \sin(\omega_\oplus T_\oplus) + A_c \cos(\omega_\oplus T_\oplus) + B_s \sin(2\omega_\oplus T_\oplus) + B_c \cos(2\omega_\oplus T_\oplus)$$

- A_s, A_c, B_s, and B_c are dependent on the $\hat{N}^X$, $\hat{N}^Y$, $\hat{N}^Z$, and the relevant SME components

$$(a_L)^X,\ (a_L)^Y,\ (c_L)^{TX},$$
$$(c_L)^{TY},\ (c_L)^{XX},\ (c_L)^{XY},$$
$$(c_L)^{XZ},\ (c_L)^{YY},\ (c_L)^{YZ}$$

- We wish to constrain these coefficients by checking for event rate modulation in the first four sidereal harmonics.

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* *Kostelecký and Mewes, Phys. Rev. D* **69**, *016005 (2004), page 2.*